RecyclerView

Latest Items



Namkeen

500gm

Rs.350

0



Sona Namkeen

250gm

Rs.100

0



Chips

500gm

Rs.80

0



Lays

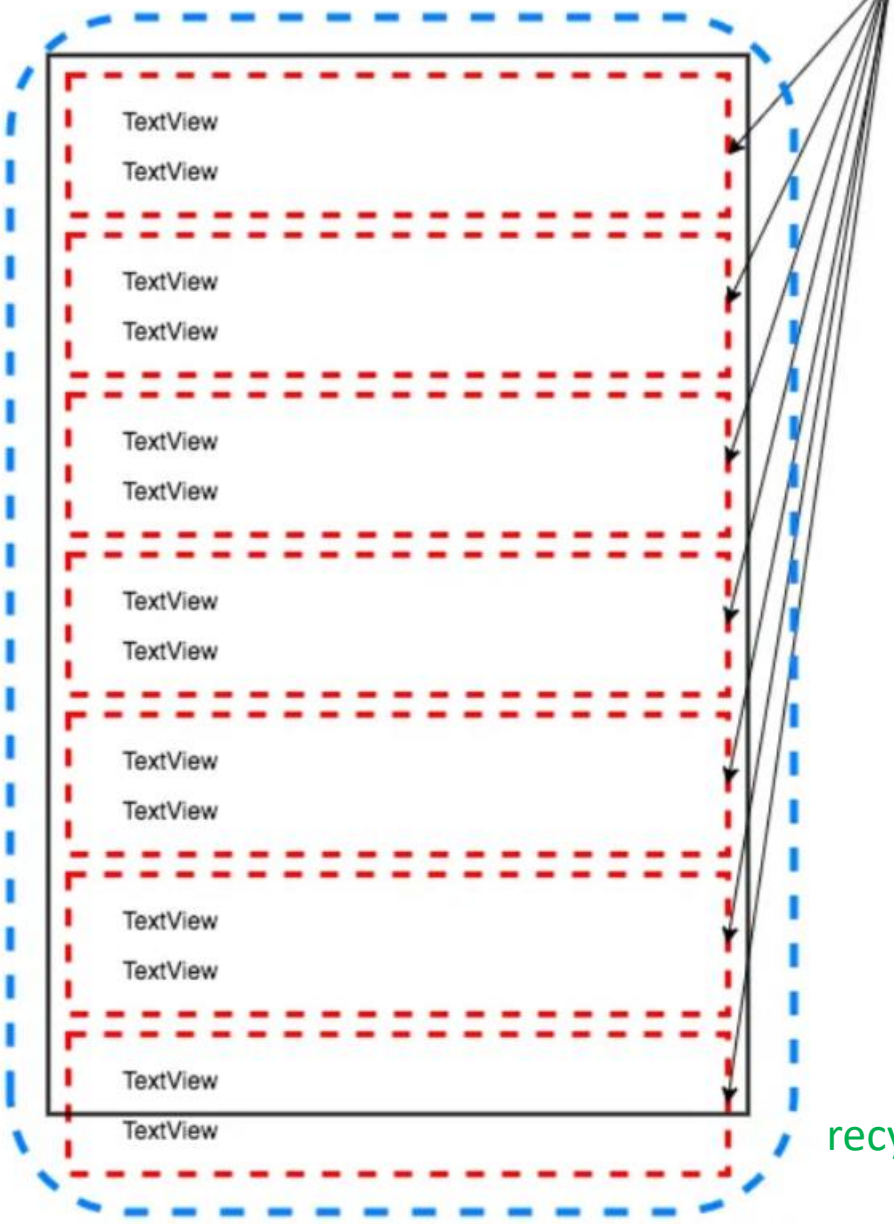
100

Rs.40

0

RecyclerView

Recycles data within a set of ViewHolders



ViewHolders

Populates the View elements in the UI

Adapter

Binds data to ViewHolders

Data Collection

Populates the views in the ViewHolders

توصیف‌کننده view یک آیتم و سلسله‌ای
درباره آن آیتم

عمارتی بین RV و داده‌ای که RV باید
نمایش دهد.

وظیفه: ایجاد ViewHolder لازم و
اتصال ViewHolder به داده‌ها

ساختار recyclerview

- RecyclerView.Adapter
- RecyclerView.ViewHolder
- RecyclerView.LayoutManager

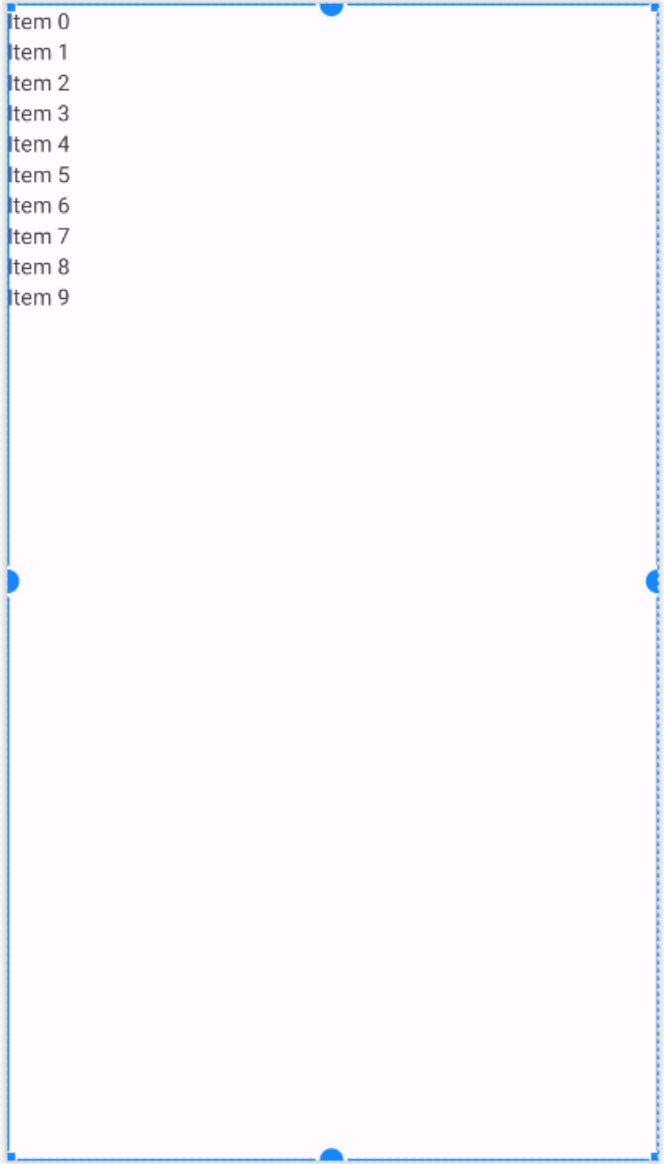
activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout xmlns:ar
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:scrollbars="vertical"
    tools:context=".MainActivity">

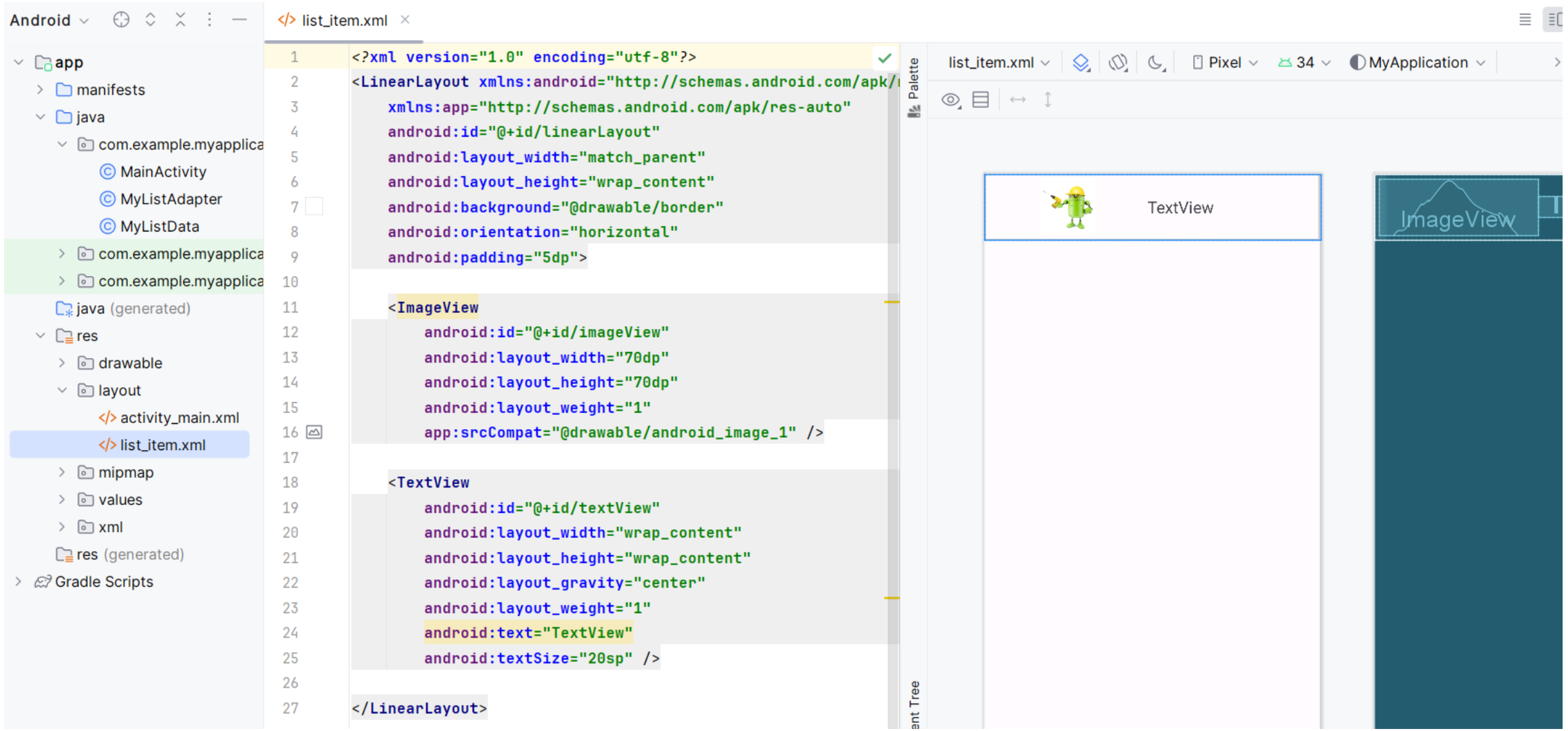
    <androidx.recyclerview.widget.RecyclerView
        android:id="@+id/recyclerView"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:scrollbars="vertical"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

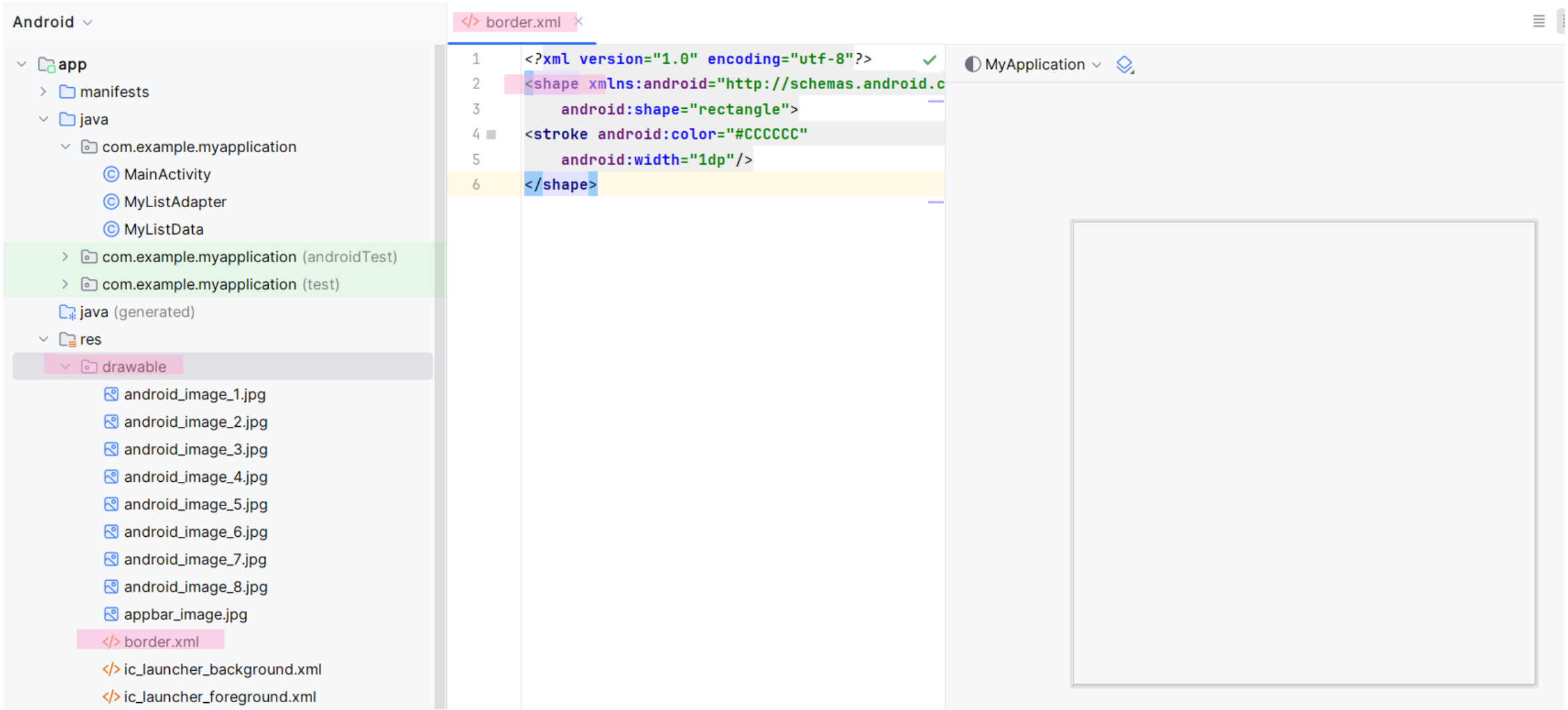
Palette

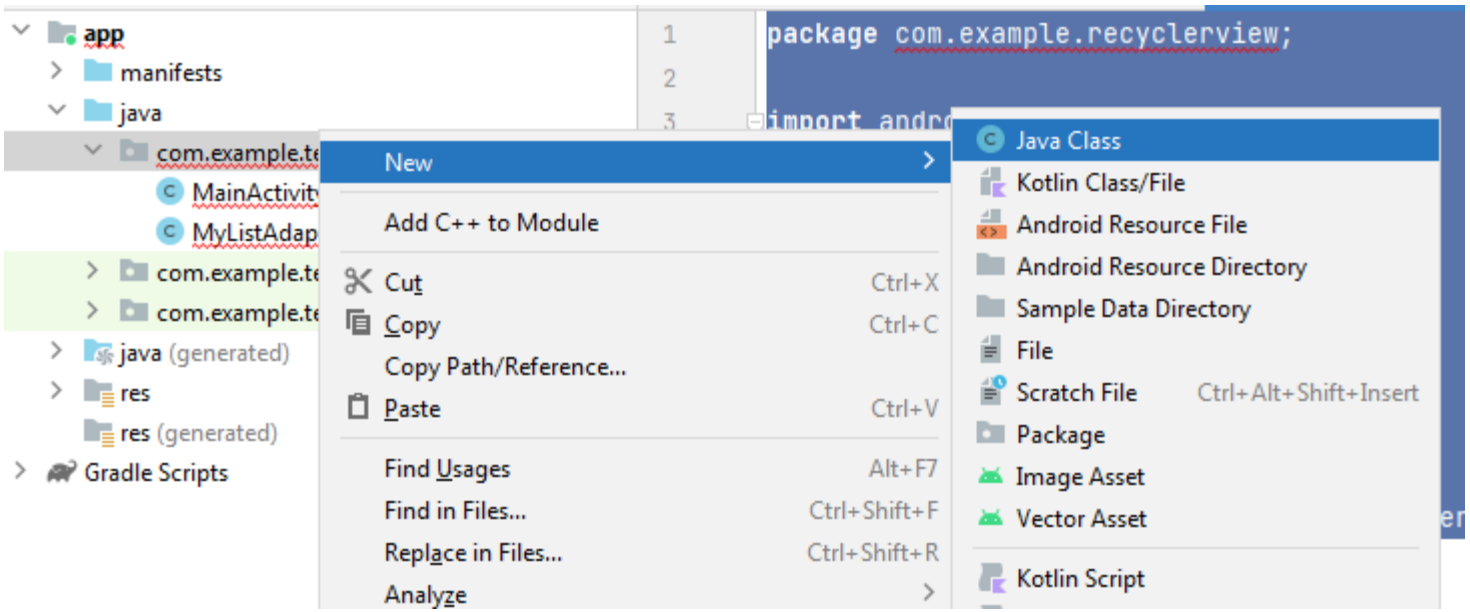
activity_main.xml | Pixel | 34 | 0dp

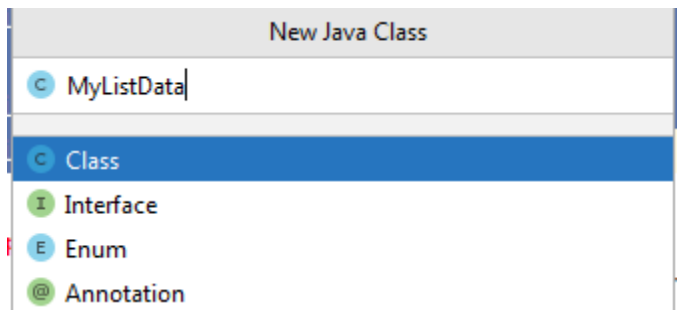


Component Tree


Android
list_item.xml
app
manifests
java
com.example.myapplica
MainActivity
MyListAdapter
MyListData
com.example.myapplica
com.example.myapplica
java (generated)
res
drawable
layout
activity_main.xml
list_item.xml
mipmap
values
xml
res (generated)
Gradle Scripts
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/
xmlns:app="http://schemas.android.com/apk/res-auto"
android:id="@+id/linearLayout"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:background="@drawable/border"
android:orientation="horizontal"
android:padding="5dp">
<ImageView
android:id="@+id/imageView"
android:layout_width="70dp"
android:layout_height="70dp"
android:layout_weight="1"
app:srcCompat="@drawable/android_image_1" />
<TextView
android:id="@+id/textView"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_gravity="center"
android:layout_weight="1"
android:text="TextView"
android:textSize="20sp" />
</LinearLayout>
Palette
list_item.xml
Pixel
34
MyApplication
TextView
ImageView


Android
app
manifests
java
com.example.myapplication
MainActivity
MyListAdapter
MyListData
com.example.myapplication (androidTest)
com.example.myapplication (test)
java (generated)
res
drawable
android_image_1.jpg
android_image_2.jpg
android_image_3.jpg
android_image_4.jpg
android_image_5.jpg
android_image_6.jpg
android_image_7.jpg
android_image_8.jpg
appbar_image.jpg
border.xml
ic_launcher_background.xml
ic_launcher_foreground.xml
border.xml
1 <?xml version="1.0" encoding="utf-8"?>
2 <shape xmlns:android="http://schemas.android.c
3 android:shape="rectangle">
4 <stroke android:color="#CCCCCC"
5 android:width="1dp"/>
6 </shape>
MyApplication


app
manifests
java
com.example.te
MainActivit
MyListAdap
com.example.te
com.example.te
java (generated)
res
res (generated)
Gradle Scripts
1
2
3
package com.example.recyclerview;
import andro
New
Add C++ to Module
Cut Ctrl+X
Copy Ctrl+C
Copy Path/Reference...
Paste Ctrl+V
Find Usages Alt+F7
Find in Files... Ctrl+Shift+F
Replace in Files... Ctrl+Shift+R
Analyze
Java Class
Kotlin Class/File
Android Resource File
Android Resource Directory
Sample Data Directory
File
Scratch File Ctrl+Alt+Shift+Insert
Package
Image Asset
Vector Asset
Kotlin Script


New Java Class
MyListData
Class
Interface
Enum
Annotation

MyListData.java

```java
package com.example.myapplication;

29 usages
public class MyListData {
    2 usages
    String title;
    2 usages
    int imgId;

    //..............Constructor: to MainActivity
    24 usages
    public MyListData(String title, int imgId) {
        this.title = title;
        this.imgId = imgId;
    }

    //..............Getters: will be called in Adapter class
    // (MyListAdapter), customize layout: List_item
    2 usages
    public String getTitle() { return title; }

    1 usage
    public int getImgId() { return imgId; }

}//MyListData
```

MyListAdapter.java

```java
package com.example.myapplication;

import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import android.widget.ImageView;
import android.widget.LinearLayout;
import android.widget.TextView;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.recyclerview.widget.RecyclerView;

3 usages
public class MyListAdapter extends RecyclerView.Adapter <MyListAdapter.ViewHolder> {

    5 usages
    private MyListData[] listData;
    //.....Constructor:
    1 usage
    public MyListAdapter(MyListData[] listData) {

        this.listData = listData;
    }
```

ایجاد کلاس **Adapter** جهت اتصال داده‌ها به آیتم‌های قابل نمایش در **RecyclerView**.

کلاس داخلی برای نگهداری View هر آیتم

MyListAdapter.java

XML → View

```
24      @Override
25      public ViewHolder onCreateViewHolder(@NonNull ViewGroup parent, int viewType) {
26          LayoutInflater layoutInflater=LayoutInflater.from(parent.getContext());
27          View listItem=layoutInflater.inflate(R.layout.list_item,parent, attachToRoot: false);
28          ViewHolder viewHolder=new ViewHolder(listItem);
29          return viewHolder;
30      }//onCreateViewHolder
31
32      @Override
33      public void onBindViewHolder(@NonNull ViewHolder holder, int position) {
34          MyListData myListData=listData[position];
35          holder.textView.setText(listData[position].getTitle());
36          holder.imageView.setImageResource(listData[position].getImgId());
37          holder.linearLayout.setOnClickListener(new View.OnClickListener() {
38              @Override
39              public void onClick(View view) {
40                  Toast.makeText(view.getContext(), text: "clicked on item: "+myListData.getTitle(), Toast.LENGTH_LONG).show();
41              }
42          });
43      }//onBindViewHolder
```

مدیریت viewHolderها از طریق:

۱- ایجاد viewHolder در صورت نیاز

۲- اتصال viewHolderها به داده‌های متناظر با در نظر گرفتن موقعیت (position)

MyListAdapter.java

```
45      @Override
46      public int getItemCount() {
47
48          return listData.length;
49      }
50
        5 usages
51      public class ViewHolder extends RecyclerView.ViewHolder {
52
            2 usages
53          public ImageView imageView;
            2 usages
54          public TextView textView;
            2 usages
55          public LinearLayout linearLayout;
56
57
            1 usage
58          public ViewHolder(@NonNull View itemView) {
59              super(itemView);
60              this.imageView=itemView.findViewById(R.id.imageView);
61              this.textView=itemView.findViewById(R.id.textView);
62              this.linearLayout=itemView.findViewById(R.id.linearLayout);
63          }//ViewHolder
64      }//class ViewHolder
65  }//MyListAdapter
```

تعداد کل مجموعه داده‌ها توسط Adapter نگه داشته می‌شود.
مورد استفاده در onBind

نمایش viewها توسط شی ViewHolder

MainActivity.java

```java
@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    MyListData[] myListData = new MyListData[] {
            new MyListData( title: "image 1",R.drawable.android_image_1),
            new MyListData( title: "image 2",R.drawable.android_image_2),
            new MyListData( title: "image 3",R.drawable.android_image_3),
            new MyListData( title: "image 4",R.drawable.android_image_4),
            new MyListData( title: "image 5",R.drawable.android_image_5),
            new MyListData( title: "image 6",R.drawable.android_image_6),
            new MyListData( title: "image 7",R.drawable.android_image_7),
            new MyListData( title: "image 8",R.drawable.android_image_8),
            new MyListData( title: "image 9",R.drawable.android_image_1),
            new MyListData( title: "image 10",R.drawable.android_image_2),
            new MyListData( title: "image 11",R.drawable.android_image_3),
            new MyListData( title: "image 12",R.drawable.android_image_4),
            new MyListData( title: "image 13",R.drawable.android_image_5),
            new MyListData( title: "image 14",R.drawable.android_image_6),
            new MyListData( title: "image 15",R.drawable.android_image_7),
            new MyListData( title: "image 16",R.drawable.android_image_8),
            new MyListData( title: "image 17",R.drawable.android_image_1),
            new MyListData( title: "image 18",R.drawable.android_image_2),
            new MyListData( title: "image 19",R.drawable.android_image_3),
            new MyListData( title: "image 20",R.drawable.android_image_4),
            new MyListData( title: "image 21",R.drawable.android_image_5),
            new MyListData( title: "image 22",R.drawable.android_image_6),
            new MyListData( title: "image 23",R.drawable.android_image_7),
            new MyListData( title: "image 24",R.drawable.android_image_8),
    };
```

```
        RecyclerView recyclerView=findViewById(R.id.recyclerView);
        MyListAdapter adapter=new MyListAdapter(myListData);
        recyclerView.setLayoutManager(new LinearLayoutManager( context: this));
        recyclerView.setAdapter(adapter);

    }//onCreate
}//MainActivity
```

تنظیم چیدمان

تنظیم adapter